SPE-229825-MS

Characterizations of the Physical and Chemical Properties of Very Low Sulfur Fuel Oil (VLSFO) from the Fujairah Port Area of the UAE

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

The ecological degradation of the marine environment associated with increased ship traffic is a growing concern. Increased marine transportation leads to a higher risk of oil and other hazardous spills in the seas and, thus, higher risks to marine and coastal environments. (Zhu et al., 2020). The ocean-going ships use different types of marine fuel oil to power their engines. Before 2020, 90% of ships used Heavy Fuel Oil (HFO) as their primary marine fuel, which was high in energy and relatively inexpensive. (Bilgili 2021). The downside of this heavy fuel was its high Sulfur emissions, with a Sulfur content exceeding 3.5%. Sulfur dioxide is an indirect greenhouse gas that can cause rapid global warming (Ward, 2009).

To support global efforts in combating climate change and reducing greenhouse gas (GHG) emissions from vessels into the marine environment, the International Maritime Organization (IMO), a specialized agency of the United Nations responsible for ensuring the safety and security of maritime navigation while preventing marine and atmospheric pollution from ships, established a global upper limit for Sulfur content in fuel oil utilized by vessels effective January 1, 2020 (Bazari, 2020). The previously utilized heavy high sulfur fuel oil (HSFO) has been replaced by a new category of marine fuel known as "Ultra Low Sulfur Oils" - ULSFO (S < 0.10 % m/m), and "Very Low Sulfur Oils" - VLSFO (S < 0.50 % m/m) (Daling and S0rheim, 2020). After the 2020 regulatory change in marine fuel, usage of low-sulfur fuel oils has experienced exponential growth across the world.

The Strait of Hormuz is the key transit point for all the major oil-producing countries in the Middle East, including Iran, Iraq, Kuwait, Saudi Arabia, the United Arab Emirates, Bahrain, Qatar, and Oman. Due to its strategic location near the Strait of Hormuz, Fujairah Port in the UAE is a significant commercial shipping hub for storing and refueling for marine transportation. After the IMO 2020 Regulation, the United Arab Emirates (UAE) also transitioned to low-sulfur fuel oil. Large quantities of VLSFO (Very Low Sulphur Fuel Oil) are stored and transported in Fujairah. The Arabian Gulf is often susceptible to oil spills from the cargo of the tankers and their fuel oil bunker. Due to the recent introduction of VLSFO to the fuel market, very little information is available regarding the properties and behavior of the VLSFO used in Fujairah port. VLSFOs are produced by blending refinery residual fuel with various chemicals to reduce sulfur content. These VLSFO blends often face issues such as sludge formation and sediment accumulation (Ju & Jeon, 2022). Prior studies on the environmental effects of marine fuel oil spills have predominantly focused on

high-sulfur fuel oil, with limited focus to explore the consequences of very low-sulfur fuel oil spills. The primary objective of this study is to characterize and benchmark the key features of VLSFO used in the bunkering facilities at Fujairah Port in the UAE, a critical oil storage and transportation hub for the country and the ships navigating this region.

Introduction

The ecological degradation of the marine environment associated with increased ship traffic is a growing concern. Increased marine transportation leads to a higher risk of oil and other hazardous spills in the seas and, thus, higher risks to marine and coastal environments. (Zhu et al., 2020). The ocean-going ships use different types of marine fuel oil to power their engines. Before 2020, 90% of ships used Heavy Fuel Oil (HFO) as their primary marine fuel, which was high in energy and relatively inexpensive. (Bilgili 2021). The downside of this heavy fuel was its high Sulfur emissions, with a Sulfur content exceeding 3.5%. Sulfur dioxide is an indirect greenhouse gas that can cause rapid global warming (Ward, 2009).

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For this study, samples from the Fujairah Port area were analyzed to understand the physical and chemical characteristics of the various types of VLSFO available and the variations between these types. Sixteen VLSFO (very low-sulfur fuel oils) samples were selected from storage facilities and ship owners within Fujairah Port for this study. This data will aid marine pollution responders in understanding the variable characteristics of this new fuel, particularly in planning and operations related to VLSFO spill accidents. It will also provide oil spill modelers with input parameters, including density, kinematic viscosity, and API gravity, for oil spill modeling tools.

Study Area and Geological Setting of the Gulf Region

The Arabian Gulf, also called the Arabian Gulf, is a shallow marginal sea with an average depth of 35 meters (Figure 1). It is classified as a semi-enclosed sea, exhibiting a maximum width of 340 kilometers. It narrows to 55 kilometers at the Strait of Hormuz, establishing its connection to the expansive Gulf of Oman (Misachi, 2021a). The arid climate prevalent in this region leads to an evaporation rate ranging from 1.4 to 2.1 meters per year (Ibrahim and Eltahir, 2019). The limited connection to the open sea, along with the arid climate, results in increased uncertainties and risks associated with the behavior of hydrocarbons following a spill within this confined Arabian Gulf basin (Guo et al., 2022). Fujairah port represents the only seaport along the eastern coastline of the United Arab Emirates. In addition to serving as a gateway for global trade originating from the Arabian Gulf, it meets the needs of the shipping industry as vessels transit through the Hormuz Strait. As a result, a substantial quantity of marine fuel is stored and transported in this area. According to S&P Global Commodity Insights, the sales of low-sulfur marine fuel at Fujairah port reached approximately 5.6 million tons in 2023. Hundreds of ships navigate Fujairah port daily. These factors heighten the Arabian Gulf's vulnerability to oil spill incidents (Issa & Vempatti, 2018).

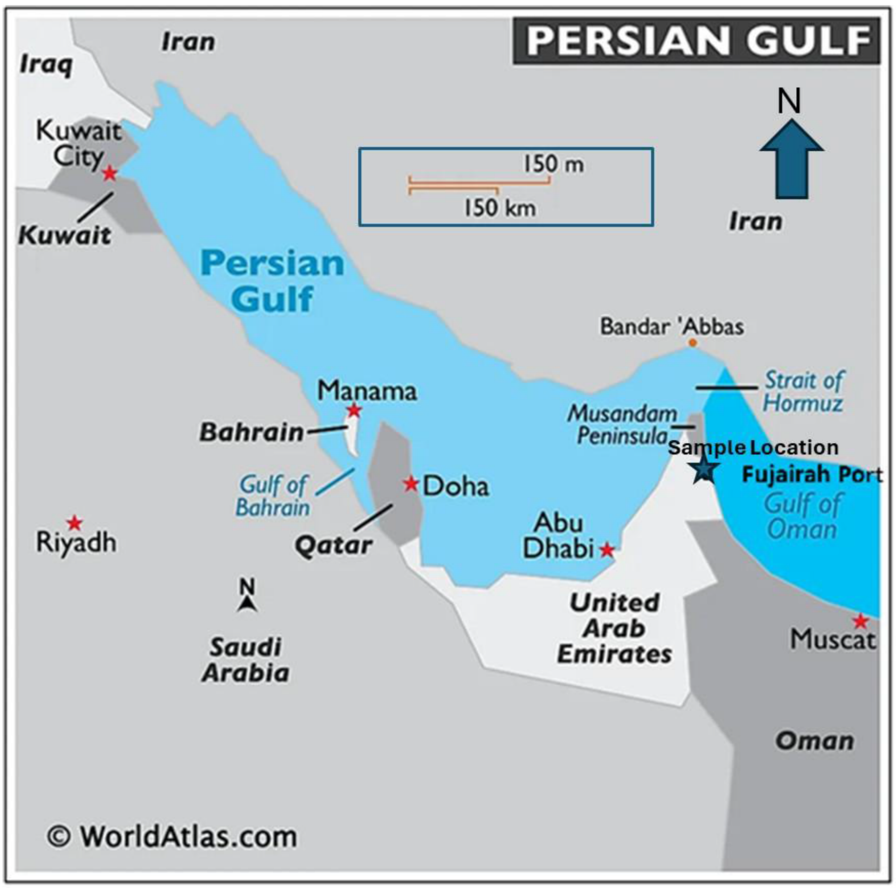


Figure 1—Arabian Gulf and Fujairah Port, (modified after Arabian Gulf World Atlas, 2021)

Sulfur Emissions by Marine Fuel

Fuels used for marine transportation include blended petroleum-based fuels, which produce SO2 emissions during oil combustion in ship engines. On a global scale, ships are responsible for approximately 5-8% of anthropogenic SO2 emissions (Corbett et al. 2007). Sulfur dioxide is an indirect greenhouse gas and can contribute to rapid global warming (Ward, 2009). SO2 emissions significantly threaten human health and the environment, resulting in acid rain and photochemical smog (Melkonyan and Krumme, 2019). Uncontrolled sulfur emissions can also severely threaten the marine ecosystem (Cullinane and Edwards, 2010). Coastal communities that depend on the sea are significantly affected by marine pollution (Smith, 2022).

Experimental setup and procedures

Recognized as a significant marine refueling facility, Fujairah Port accommodates substantial volumes of various categories of fuel oil. Sixteen VLSFO samples were collected from the Fujairah Port area and analyzed for this study. Additionally, eight HSFO and nine LSMGO (Low-Sulfur Marine Gas Oil) samples were collected to understand the differences in properties among all marine fuels used at this port. These samples were gathered from various sources. Table 1 below provides detailed information regarding all the samples collected from the study area.

Table 1—Sample and Vessel Details

SL No	Sample Number	Oil Name	Sample Source
1	VS-1	VLSFO	Chemical Tanker
2	VS-2	VLSFO	Oil Product Tanker
3	VS-3	VLSFO	Oil/ Chemical Tanker
4	VS-4	VLSFO	Oil Product Tanker
5	VS-5	VLSFO	Oil Product Tanker
6	VS-6	VLSFO	Oil Product Tanker
7	VS-7	VLSFO	Oil Product Tanker
8	VS-8	VLSFO	Oil Product Tanker
9	VS-9	VLSFO	Bunker
10	VS-10	VLSFO	Bunker
11	VS-11	VLSFO	Bunker
14	VS-14	VLSFO	Bunker
15	VS-15	VLSFO	Bunker
16	VS-16	VLSFO	Bunker
12	VS-12	VLSFO	Refinery
13	VS-13	VLSFO	Refinery
17	HS 1	HSFO	Oil Product Tanker
18	HS 2	HSFO	Oil Product Tanker
19	HS 3	HSFO	Oil Product Tanker
20	HS 4	HSFO	Oil Product Tanker
21	HS 5	HSFO	Oil Product Tanker
22	HS 6	HSFO	Oil Product Tanker
23	HS 7	HSFO	Bunker
24	HS 8	HSFO	Bunker
25	LS 1	LSMGO	Bunker
26	LS 2	LSMGO	Bunker
27	LS 3	LSMGO	Chemical Tanker
28	LS 4	LSMGO	Chemical Tanker
29	LS 5	LSMGO	Chemical Tanker
30	LS 6	LSMGO	Chemical Tanker
31	LS 7	LSMGO	Chemical Tanker
32	LS 8	LSMGO	Chemical Tanker
33	LS 9	LSMGO	Multi Purpose Offshore Vessel

VLSFO Samples

Three types of VLSFO Samples are analyzed for this study. They are Bunker Samples—In the shipping industry, bunkering refers to the provision of marine fuel to vessels. Typically, ships cannot carry the substantial amount of fuel required for extensive voyages; thus, bunkering services supply the necessary fuel to ships at various ports. Six representative Very Low Sulfur Fuel Oil (VLSFO) samples, ranging in volume from 1 to 2 liters, were collected from the Fujairah Bunker Storage Facilities. Specialized hoses are used to obtain these samples. The hoses are attached to the sample container, which, once filled, is sealed and labeled with the corresponding names of the fuels sampled. Refinery Storage Samples—Certificate of Quality (COA) of two VLSFO samples from two different refinery storage facilities near the study area are evaluated for this study.

Ship Samples—Eight samples weighing 1-2 litres are collected from different marine vessels in the Fujairah Port. These samples were collected directly from the ship's tanks using drip sampling, a method that involves collecting the sample in a container as it drips from a sampling line during the ship's fueling process.

Parameters Evaluated

All the samples collected (VLSFO, HSFO, and LSMGO) were provided to two oil testing laboratories in the UAE to perform sulfur content tests and analyze physical properties such as Kinematic Viscosity, Density, Pour Point, and Hydrocarbon content. These properties and the testing methods are briefly discussed below.

Kinematic Viscosity - The kinematic viscosity of marine fuel is critical in understanding its fluidity and also has significant implications for marine pollution at various temperatures. High-viscosity fuel can lead to the deposition of sludge and soot, which contributes to pollution; additionally, viscous oil are harder to manage, recover and dispose.

Marine fuel viscosity is typically expressed as kinematic viscosity and usually measured in centistokes (cSt), where one cSt = 1 mm^2/s. The reference temperature for the kinematic viscosity of residual fuel, VSLFO, is 50 °C, whereas for distillate fuels, a reference temperature of 40 °C is specified due to their lighter components. Both the ASTM D445 and ISO 3014 methods use glass capillary kinematic viscometers to determine the viscosity, v, of the fuel.

Density -Density measurements represent a significant characteristic of marine fuel. Elevated fuel density, particularly in the case of diesel, is correlated with increased emissions of pollutants such as nitrogen oxides (NOx) and particulate matter (PM). This phenomenon occurs because fuels with higher densities may lead to diminished engine efficiency and alterations in combustion processes, ultimately resulting in heightened emissions. The density of marine fuel is typically expressed in g/mL (g/cm^3) or kg/m^3 at 15 °C.

Pour Point -The lowest temperature at which a fuel maintains its flow solely under the influence of gravity is referred to as the pour point. The pour point does influence the handling and usage of fuel, which can have indirect implications for spill pollution response, particularly in colder climates. Products with pour points above ambient temperature require heating to facilitate their handling. For instance, a high pour point may result in the clogging of fuel filters or stalling of engines, which can potentially lead to increased fuel consumption or inefficient operation, thereby indirectly contributing to elevated emissions. Furthermore, fuels with low pour points are often essential for operations in cold weather, and the additives utilized to reduce pour point may carry environmental consequences.

The International Maritime Organization (IMO) has established 30 °C as the maximum permissible pour point limit for VLSFO fuels. If the temperature exceeds this threshold, the VLSFO will solidify, necessitating additional energy input to restore its flow characteristics. Pour points are typically measured in degrees Celsius (°C) but may also be expressed in degrees Fahrenheit in specific contexts.

Sulfur Levels - Crude oils and their refined products, including residual and distillate marine fuels, may contain various inorganic and organic sulfur compounds. As previously indicated, sulfur emissions significantly harm human health and the environment. The sulfur content of very low sulfur fuel oil (VLSFO) must not exceed 0.5%, according to International Maritime Organization (IMO) regulations. The unit for measuring sulfur is expressed in parts per million (ppm) or as a percentage by mass.

SARA (Saturates, Asphaltenes, Resins, and Aromatics) - The structural composition of a hydrocarbon fuel is determined by its saturates (SAT), aromatics (ARO), resins (RES), and asphaltenes (ASP), collectively referred to as SARA. The SARA components are expressed as % weight. Residual Marine Fuel, which is the heavy residue resulting from refinery processing, may contain natural surfactants, including asphaltenes and resins, that typically form an emulsion if spilled in the marine environment. SARA Analysis does not measure pollution levels; however, it provides valuable information for understanding the potential for pollution if spilled at sea. Having such information allows spill scientists to more accurately predict the behavior of pollutants in the environment, which is essential for efforts aimed at pollution prevention and remediation.

Oil Spill Analysis - One of the key parameters for oil spill responders is understanding the viscosity of weathered VLSFO residues. Any changes in the viscosity of the weathered oil residues become a crucial factor for spill responders, as this can impact skimming, pumping, and oil transfer equipment during recovery at sea. Increases in viscosity can also significantly hinder the effectiveness of skimmers and chemical dispersants.

The spill simulations are usually performed by state-of-the-art software like RPS's Oil Spill Modeling framework (OILMAP/SIMAP). Each oil spill event is simulated using the stochastic model to evaluate the footprint and likelihood of marine/coastal areas to be polluted, and the deterministic model to quantify the worst-case scenario that will lead to the largest volume of hydrocarbon reaching shore in the shortest time.

Testing Methods

Two standard methods for testing Marine Fuel Oil are used to ensure that the properties of marine fuels are within the MARPOL Annex VI standard (US EPA, 2013). These are

1. The ASTM (American Society for Testing and Materials) (ASTM) Standards, n.d.)
2. ISO (International Organization for Standardization) (Transportpolicy.net, 2018)

Both the ASTM and ISO standards are common methods that are equivalent to each other. The ISO standards are usually used to test the comprehensive marine fuel quality, whereas the ASTM standards are used for detailed testing of the individual properties of the fuels. (Lidia, 2023)

As of 1 January 2020, certain specifications for low-sulfur residual marine fuels have been introduced within ISO 8217 to minimize the risk of incompatibility between these "new" fuels, primarily for the shipping industry (IMO, 2019).

Results

Kinematic Viscosity variation of the VLSFO samples

The viscosity of marine fuel does not serve as a direct indicator of its quality; rather, the refining process can be discerned through viscosity measurements. Historically, marine fuels were classified based on their viscosity and density; however, following the IMO 2020, sulfur content has emerged as the predominant criterion for the classification of marine fuels. Table 2 lists the kinematic viscosity (cSt) of the VLSFO samples at the designated temperature of 50 °C. A wide range of viscosity values is observed, with the lowest value of 131.4 cSt and the highest value of 372.3 cSt. Although there is no significant relationship between viscosity and sample source, the refinery samples do have lower viscosity than the other samples collected. Refineries mostly use a complex refining process to process different types of crude oil from various sources (Vermeire, 2021). The complex refining has a definite impact on the characteristics of the marine fuels. More refinery samples will be required to understand the relationship between the source and its viscosity.

Table 2—VLSFO Kinematic Viscosity at 50 °C

Sample Number	Kinematic Visc (cSt) at 50oC	Visc Methodology	Sample Source
VS-1	131.4	ISO 3104	Chemical Tanker
VS-2	152.3	ASTM D445	Oil Product Tanker
VS-3	236.4	ASTM D 445	Oil/ Chemical Tanker
VS-4	297.8	ISO 3104	Oil Product Tanker
VS-5	211.6	ISO 3104:2023	Oil Product Tanker
VS-6	372.2	ISO 3104:2023	Oil Product Tanker
VS-7	207.1	ISO 3104:2023	Oil Product Tanker
VS-8	148.8	ASTM D 445	Oil Product Tanker
VS-9	135.4	ASTM D 445	Bunker
VS-10	156.1	ASTM D 445	Bunker
VS-11	135.5	ASTM D 445	Bunker
VS-12	141.3	ISO 3104	Refinery
VS-13	125.8	ISO 3104	Refinery
VS-14	317	ASTM D 445	Bunker
VS-15	167.4	ISO 3104	Bunker
VS-16	210.4	ISO 3104	Bunker
Mean	196.66		
Standard Deviation (SD)	72.55		
Standard Error (SE)	18.14		

Density values of the VLSFO samples

The density and API gravity of the 16 VLSFO samples are presented in Table 3. The maximum density among the 16 VLSFO samples was recorded at 0.9467 kg/L, while the minimum was noted as 0.9188 kg/L, resulting in an average density of 0.9285 kg/L. Based on API gravity classifications, fourteen of the VLSFO samples would be categorized as "heavy oils," whereas two VLSFO samples would be designated as "medium oils."

Table 3—VLSFO Density and API Oil Classification

Sample Number	Density (Kg/l) at 15oC	Density Methodology	API Classificat ion	Sample Source
VS-1	0.9223	ISO 3675	Heavy	Chemical Tanker
VS-2	0.9252	ISO 3675	Heavy	Oil Product Tanker
VS-3	0.9329	ASTM D 4294	Heavy	Oil/ Chemical Tanker
VS-4	0.9246	ASTM D 4294	Heavy	Oil Product Tanker
VS-5	0.937	ISO 12185:2024	Heavy	Oil Product Tanker
VS-6	0.9467	ISO 12185:2024	Heavy	Oil Product Tanker
VS-7	0.9235	ISO 12185:2024	Heavy	Oil Product Tanker
VS-8	0.931	ASTM D1298	Heavy	Oil Product Tanker
VS-9	0.9188	ASTM D1298	Medium	Bunker
VS-10	0.9224	ASTM D1298	Heavy	Bunker
VS-11	0.9195	ASTM D1298	Medium	Bunker
VS-12	0.9261	ISO 12185	Heavy	Refinery
VS-13	0.9385	ISO 12185	Heavy	Refinery
VS-14	0.9295	IP 365	Heavy	Bunker
VS-15	0.9295	ISO 12185:2024	Heavy	Bunker
VS-16	0.9282	ISO 12185:2024	Heavy	Bunker
Mean	**0.9284813**			
Standard Deviation (SD)	**0.0074779**			
Standard Error(SE)	**0.0018695**			

Relationship between Viscosity and Density

Although viscosity and density exhibit distinct characteristics, both represent significant properties of fuel oils and are critical parameters for managing oil spills within this context. Table 4 lists the range of viscosity and density of the HSFO and LSMGO samples collected from the study area. Industry standards, such as those from ASTM (American Society for Testing and Materials) and ISO (International Organization for Standardization), described above, specify 40°C as a reference temperature for measuring the viscosity of LSMGO and 50°C for the kinetic viscosity of VLSFO and HSFO (Wearcheck.ca, 2017). This helps ensure consistency in testing and reporting.

Table 4—Density and viscosity of the HSFO and LSMGO samples

Sample No	Oil Name	Kinematic Visc (cSt)	Density (Kg/l)	Sample source
HS 1	HSFO	270.8	0.963	Oil Product Tanker
HS 2	HSFO	376.4	0.9652	Oil Product Tanker
HS 3	HSFO	331.8	0.974	Oil Product Tanker
HS 4	HSFO	216.6	0.9769	Oil Product Tanker
HS 5	HSFO	360.6	0.9658	Oil Product Tanker
HS 6	HSFO	310.4	0.9735	Oil Product Tanker
HS 7	HSFO	485.5	0.9947	Bunker
HS 8	HSFO	435.2	0.9936	Bunker
LS 1	LSMGO	2.681	0.9252	Bunker
LS 2	LSMGO	3.101	0.8339	Bunker
LS 3	LSMGO	2.426	0.8274	Oil Product Tanker
LS 4	LSMGO	2.567	0.8252	Oil Product Tanker
LS 5	LSMGO	3.374	0.8288	Oil Product Tanker
LS 6	LSMGO	2.703	0.8338	Oil Product Tanker
LS 7	LSMGO	2.189	0.8203	Oil Product Tanker
LS 8	LSMGO	3.143	0.8195	Oil Product Tanker
LS 9	LSMGO	2.893	0.8213	Oil Product Tanker
LS 10	LSMGO	2.891	0.824	Oil Product Tanker

The Low Sulfur Marine Gas Oil (LSMGO) samples exhibit very low viscosity and density, whereas the High Sulfur Fuel Oil (HSFO) samples are characterized by high density and viscosity. The Very Low Sulfur Fuel Oil (VLSFO) occupies an intermediary position between these two fuel types (Figure 2).

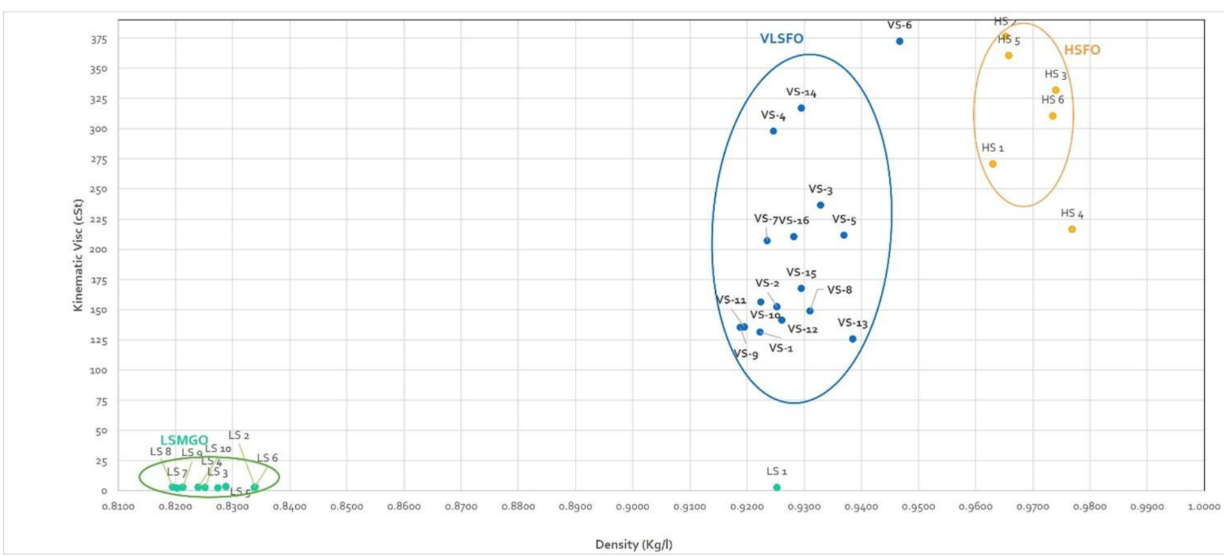


Figure 2—Density and kinematic viscosity of 33 Samples

Viscosity and Density of the VLSFO Samples

Typically, the viscosity of the Very Low Sulfur Fuel Oil (VLSFO) samples is anticipated to increase in conjunction with rising density. A correlation between the density and viscosity in the VLSFO samples is not observed (Figure 3). Analyzing the variations in viscosity that correspond to changes in density poses challenges, as multiple factors, including the composition of the crude oil, sample preservation methods, and laboratory measurement techniques, influence these values.

Figure 3—Density and viscosity in VLSFO samples from three sources.

Pour Point

A broad range of pour points is observed, with the highest at 30 °C and the lowest at -9 °C (Figure 4). VLSFO fuels have high paraffinic content, and the pour point may range from 6 °C to as high as 30 °C. Some of these samples display pour points similar to those of the LSMGO fuels, which are typically below 0 °C (Ship & Bunker, 2020).

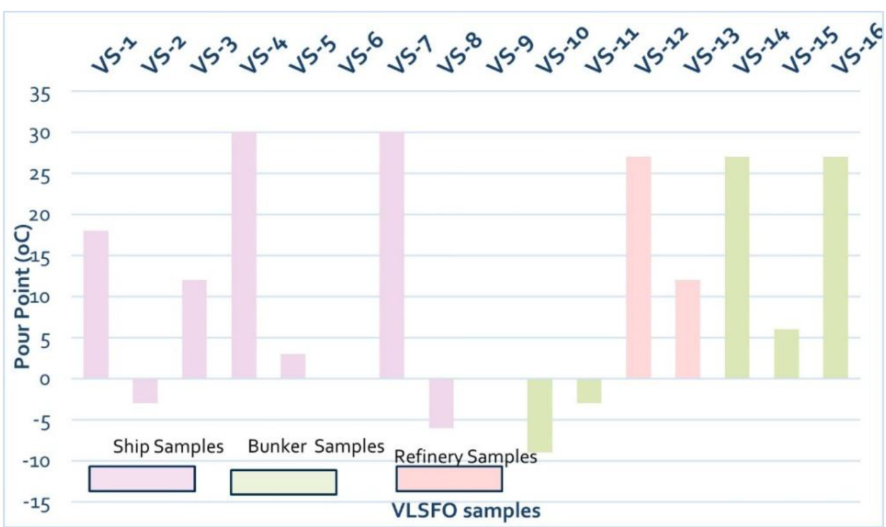


Figure 4—VLSFO Pour Point values

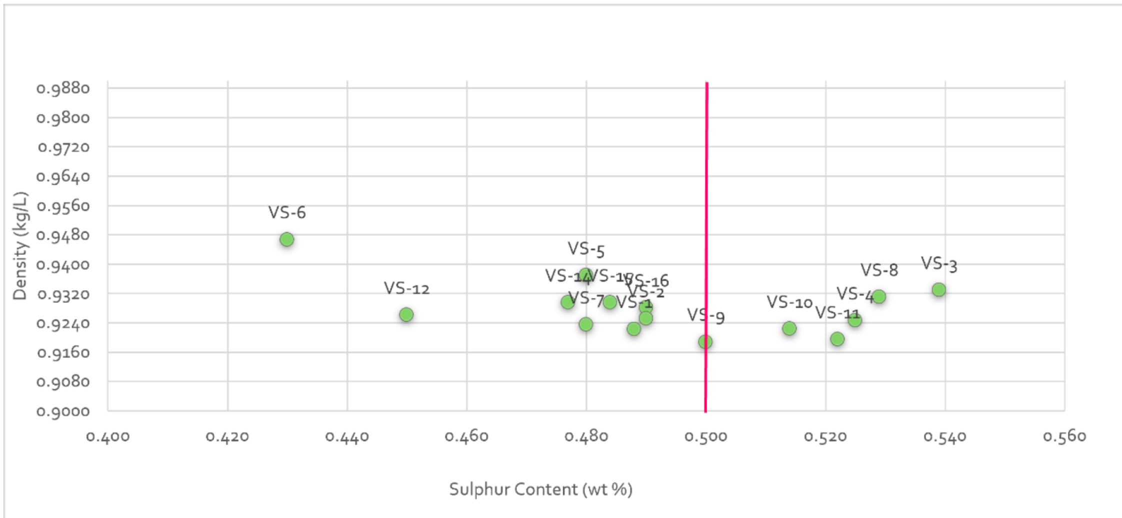
Sulfur

The highest sulfur percentage is 0.539% and the lowest is 0.43%. It is noted that five of the samples analyzed contained a higher sulfur level than the IMO 2020 directive for sulfur in VLSFO (Table 5).

Table 5—VLSFO Sulfur percentage (samples exceeding the IMO 0.5% limit are highlighted in red)

Sample Number	Sulfur wt %	Laboratory Method	Sample Source
VS-1	0.488	ISO 8754	Chemical Tanker
VS-2	0.490	ASTM D4294	Oil Product Tanker
VS-3	0.539	ASTM D4294	Oil/ Chemical Tanker
VS-4	0.525	ASTM D4294	Oil Product Tanker
VS-5	0.48	ISO 8754:2003	Oil Product Tanker
VS-6	0.43	ISO 8754:2003	Oil Product Tanker
VS-7	0.48	ISO 8754:2003	Oil Product Tanker
VS-8	0.529	ASTM D4294	Oil Product Tanker
VS-9	0.5	ASTM D4294	Bunker
VS-10	0.514	ASTM D4294	Bunker
VS-11	0.522	ASTM D4294	Bunker
VS-12	0.450	ASTM D4294	Refinery
VS-13	0.390	ASTM D4294	Refinery
VS-14	0.477	ASTM D4294	Bunker
VS-15	0.484	ISO 8754	Bunker
VS-16	0.49	ISO 8754:2003	Bunker
Mean	0.487		
Standard Deviation (SD)	0.039		
Standard Error (SE)	0.010		

VLSFO fuel exhibits reduced density and viscosity due to incorporating lighter and less viscous stocks blended with the residual fuel, aimed at minimizing sulfur content. The analysis of sulfur content in the VLSFO samples, when plotted against kinematic viscosity and density values, indicates no correlation between variations in density and viscosity with sulfur levels (Figures 5a and 5b).



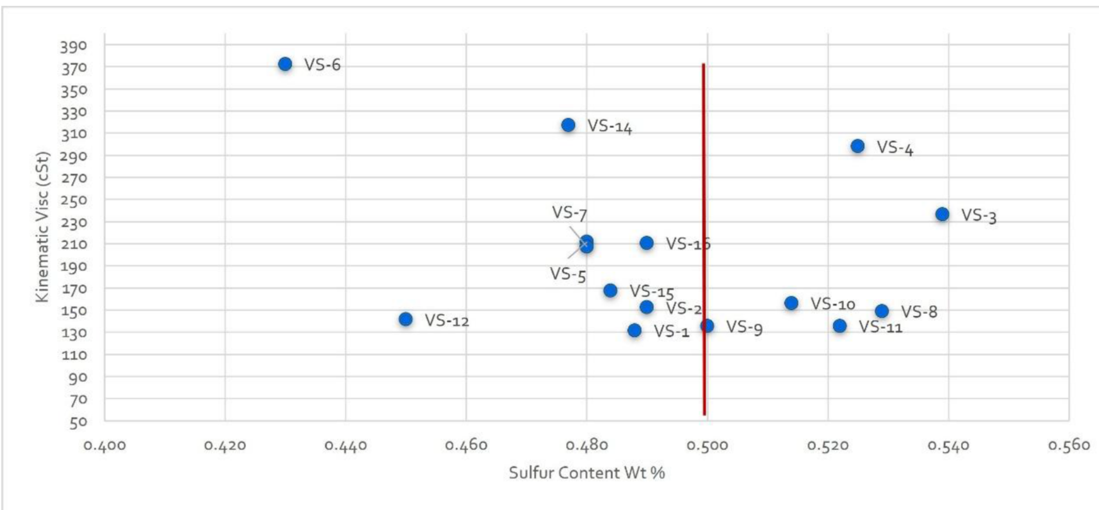


Figure 5—(Part 1) Density variation of the VLSFO samples with sulfur content, 5b (Part 2) Viscosity variation of the VLSFO samples with sulfur content (red line shows the maximum sulfur permissible by IMO of 0.5% in the VLSFO)

Kinematic Viscosity variation with temperature

The viscosity of any petroleum fuel is highly dependent on the temperature. It increases significantly as the temperature decreases. Nine VLFSO samples were tested for low-temperature kinematic viscosity at 5 °C, 15 °C, 25 °C, and 40 °C, and the results are presented in Figure 6. Two of the samples studied exhibit an exponential increase in viscosity at low temperatures. The data for the seven other samples was not used as the viscosity measurements at the lower temperatures for these samples were not very reliable.

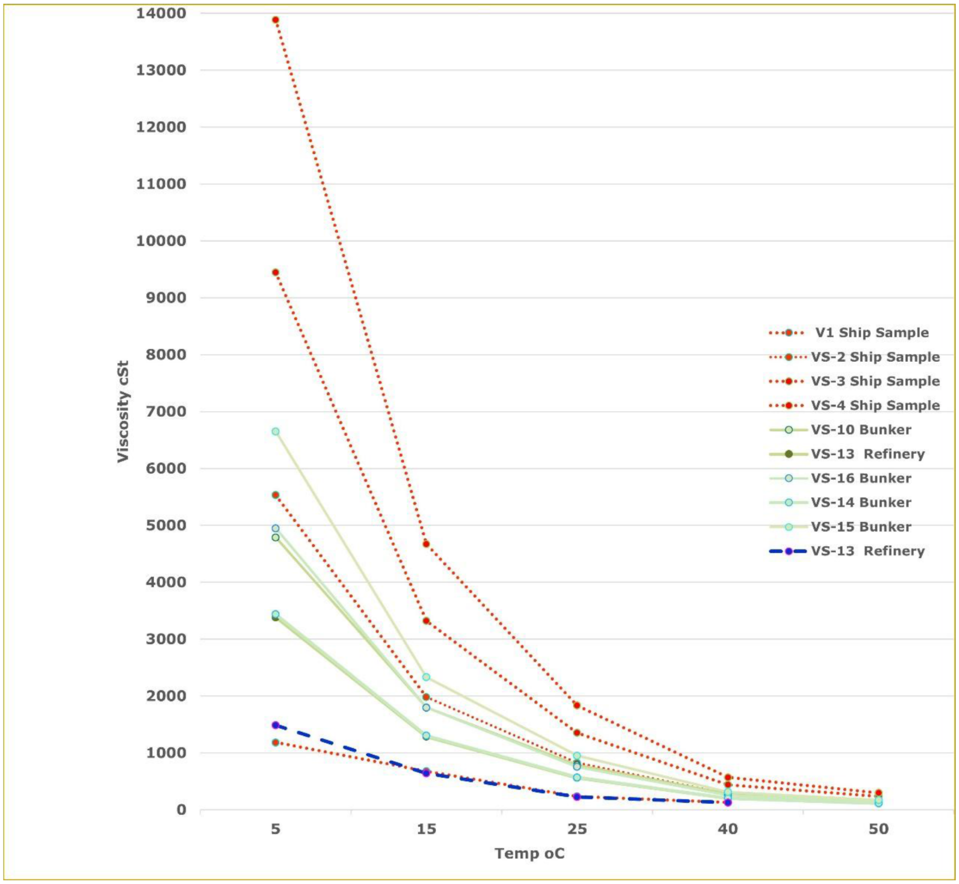


Figure 6—VLSFO Viscosity at Various Temperatures for 9 VLSFO Samples

SARA (Sulfur Aromatics Resin and Asphaltene) Analysis

Six VLSFO samples and two HSFO samples were tested for SARA Analysis (Table 6). The percentages of saturates, aromatics, resins, and asphaltenes present in the VLSFOs will facilitate an understanding of their emulsification properties in the event of a spill into marine environments.

Table 6—SARA Content for eight samples by Percentage Mass

Sample	Saturates % wt	Density (Kg/l) at 15oC	Kinematic Visc (cSt) at 50oC	Sources
VS-9	33.85	0.9188	135.4	Bunker
VS-11	36.65	0.9195	135.5	Bunker
VS-8	37.25	0.931	148.8	Oil Product Tanker
VS-4	38.08	0.9246	297.8	Oil Product Tanker
VS-10	39.12	0.9224	156.1	Bunker
VS-3	43.01	0.9329	236.4	Oil/ Chemical Tanker
HS 7	35.05	0.9947	485.5	Bunker
HS 8	38.12	0.9936	435.2	Bunker

The content of saturates represents one of the principal modifications in VLSFOs when compared to the heavy marine fuels previously utilized by maritime vessels. Due to the paraffinic characteristics of saturates, an increase in their concentration results in greater wax precipitation from the VLSFOs (Thobejane and de Vaal, 2023). The histogram (Figure 7) below illustrates the variations observed in the SARA analysis of all eight samples.

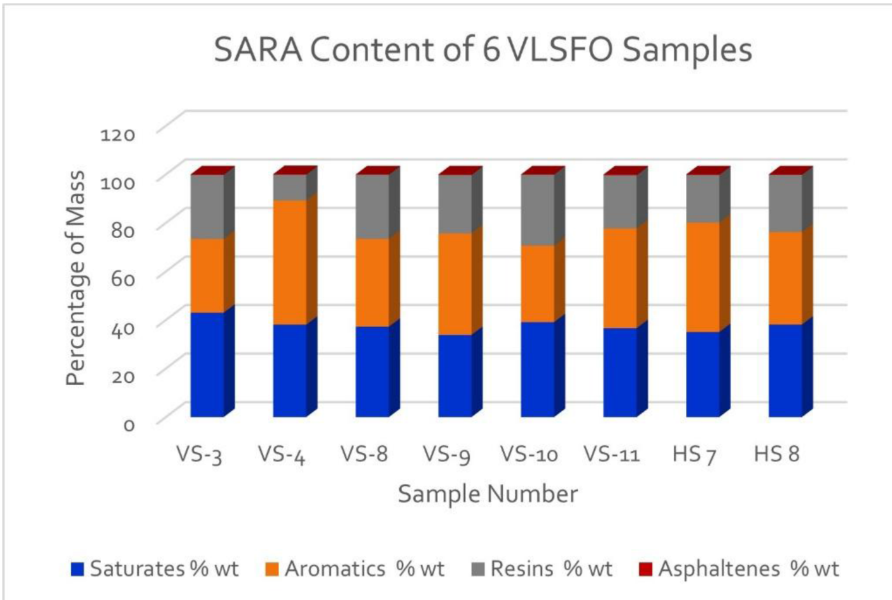


Figure 7—SARA Content for 8 VLSFOs and HSFOs by Percentage Mass

The average saturate content of all the VLSFO and HSFO samples is approximately 37.6%. Although the refining process of the VLSFO and HSFO differs, the saturate content of both fuels is almost within the same range. There is not much variation in the aromatic content of all the samples studied, except for Sample VS 4, which is 51.2%. A high aromatic content is associated with poor fuel quality, producing ultrafine particulate matter. The resin content ranges from 29.3% to 10.6%. The stability of the fuel depends on the resin content, which also helps prevent the separation of the asphaltene content of the marine fuel (Speight, 2004). The average resin content of all the samples is 22.7%, except for Sample VS 4, which is only 10.6%, thus increasing the risk to fuel stability. The Box whisker plot of the SARA Analysis (Figure 8) shows that Aromatics and the Resins have significant outliers.

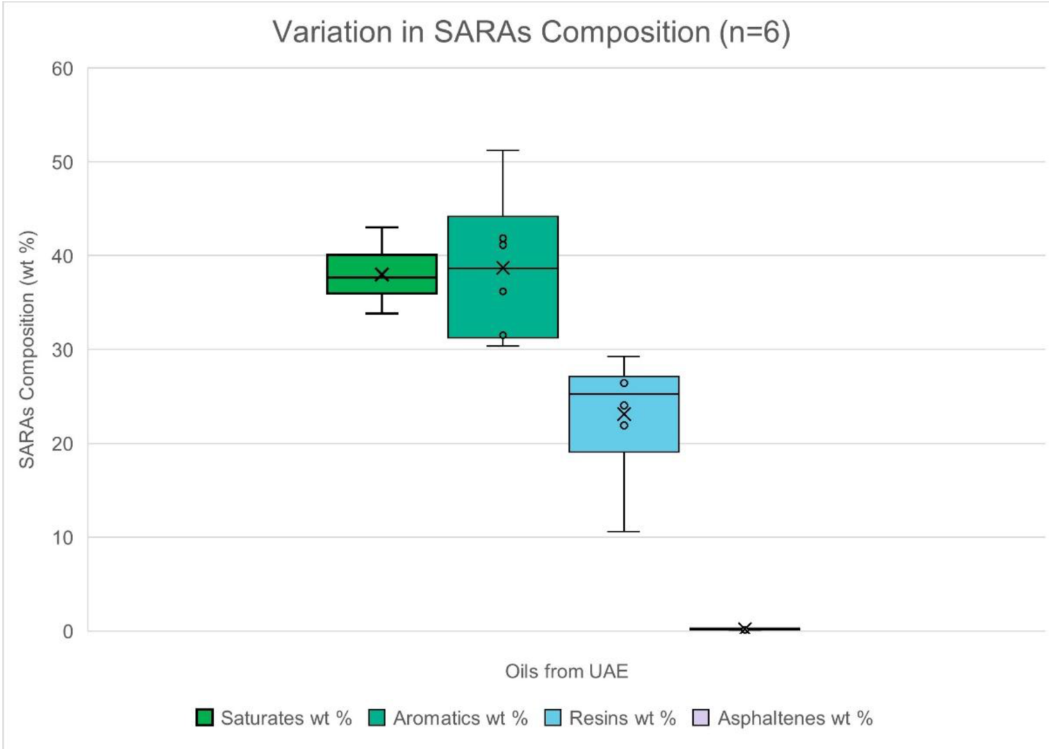


Figure 8—Variation in SARA Composition of the VLSFO Samples

VLSFO samples: Saturate content variation with viscosity and density

Saturates, non-aromatic hydrocarbons, are less dense and have lower viscosity than aromatics and resins, as seen in heavier fuel oils. A higher saturated content in a fuel will lower overall viscosity and density, making the fuel flow more easily. This relationship is not seen in the VLSFO samples tested for SARA Analysis. Table 7 shows the saturate content of six VLSFO Samples from the study area. Sample VS-3 has the highest saturate content, but the viscosity and density of these samples are also high (Table 7).

Table 7—VLSFO and HSFO Saturate Content, Density, and Viscosity

Sample	Saturates % wt	Density (Kg/l) at 15oC	Kinematic Visc (cSt) at 50oC	Sources
VS-9	33.85	0.9188	135.4	Bunker
VS-11	36.65	0.9195	135.5	Bunker
VS-8	37.25	0.931	148.8	Oil Product Tanker
VS-4	38.08	0.9246	297.8	Oil Product Tanker
VS-10	39.12	0.9224	156.1	Bunker
VS-3	43.01	0.9329	236.4	Oil/ Chemical Tanker
HS 7	35.05	0.9947	485.5	Bunker
HS 8	38.12	0.9936	435.2	Bunker

Spill Model Data Sets

To understand the characteristics of the VLSFO sample if spilled in the Arabian Gulf, an example simulation is presented from another study under similar conditions in this region, using the OILMAP (RPS, n.d.) software. These conditions are tropical sea temperatures of 20°C, 23°C, and 26°C, following an instantaneous spill of 1000 barrels (155.23 metric tons) at a constant wind speed of 4 m/s. The amount of VLSFO that might evaporate, remain, and be dispersed after a spill is summarized in Table 8 below.

Table 8—Mass Balance weathering model

Sea Surface Temperature (SST)	Sea Surface Wind Speed(m/s)	Average VLSFO Percentage Surface%	Average VLSFO Percentage Degraded %
20oC	4	98.78	1.22
23oC	4	98.78	1.22
26oC	4	98.78	1.22

Under these weather conditions, temperatures, and winds, 98% of the VLSFO will remain on the sea surface after the spill event, and only 2% of the VLSFO will be degraded, necessitating the recovery of the remaining fuel in the sea. Zero value was assigned to the percentage evaporated, dissolved, entrained, and moved ashore. However, this result cannot be considered conclusive. The prediction of an oil spill event depends on several interrelated factors, including sea conditions, meteorological data, and additional physicochemical properties of the VLSFO, which have not been obtained for this study.

VLSFO studies in other regions

Despite the growing interest in using VLSFO, there is insufficient focus on the quality and variability of this fuel. Few studies have attempted to understand the behavior of VLSFO in different environments. The Norwegian Coastal Administration (NCA, Kystverket) engaged SINTEF, a European Independent Research Group, for a couple of studies to map the relevant properties of low-sulfur oil. The results of different tests showed that low-sulfur oils exhibit different hydrocarbon profiles (gas chromatography) reflecting variations in their physicochemical properties (Daling et al., 2018).

Low sulfur fuel spill recovery testing in icy conditions in the Xamk oil spill response test basin by IMAROS (Improving response capacities and understanding the environmental impacts of new generation low sulfur Marine fuel Oil Spills). IMAROS is an EU-funded project coordinated by the Norwegian Coastal Administration with partners from Belgium, Denmark, France, Malta, and Sweden (Europa.eu, 2020). It was noted that when VLSFO spills into the ice, it begins to solidify, making recovery difficult. (Kaakkois-Suomen ammattikorkeakoulu Xamk, 2025).

Another project by the Australian Maritime Safety Authority (AMSA) under its National Plan in 2020 was completed to understand and address emergency response planning, mitigation, and safety concerns related to VLSFO marine spills. 49 VLSFO samples acquired globally were analyzed for this study. This study concluded that VLSFO samples from different bunkering ports exhibit high variability. VLSFO RMG380 samples exhibited significant variability in their physical properties, including kinematic viscosity, density, pour points, and hydrocarbon composition. The asphaltene content of the 21 VLSFOs tested varied significantly, ranging from a high of 12.5% to a low of 0.2%, indicating the aromatic nature of the source of these VLSFOs. Some samples did not meet the IMO 2020 Sulfur criteria of 0.05 %(Gilbert, 2023).

Discussion

The 16 VLSFO samples used in this study were sourced from various ship types and bunker and refinery storage facilities in the Fujairah region. The samples exhibited significant variability in hydrocarbon composition and physical properties, with some being "out of specification." Determining whether this discrepancy is due to sample contamination or a variation in laboratory measurements is challenging. This may be due to the blending of high-density, heavy fuel oils with small amounts of light oil, such as marine gas oil (MGO), to reduce viscosity and meet the specifications for low sulfur content.

The average VLSFO sample density is 0.9281 kg/l, which is lower than the average oil density of the HSFO samples analyzed. Though the oil characteristics may change after it spills into the sea, as the average density of the seawater in the Arabian Gulf is around 1.027 g/ml (M. Payehghadr and A. Eliasi, 2010), any spills of VLSFO may remain floating on the sea surface.

Nearly half of the studied VLSFO samples have high viscosities. Responding to oil spills can be difficult when sea temperatures are low, as fuel oil may solidify or too viscous to recover or skim under these conditions. Once spilled, the oil is influenced by various factors in the sea, which can lead to significant variability in its physical state and behavior.

Summary and Conclusion

The new regulation, implemented by the International Maritime Organization (IMO), was enacted in January 2020. This change in the sulfur content of marine fuels will lower sulfur emissions from marine transportation, immensely benefiting the coastal population and marine ecology.

The sixteen VLSFO samples analyzed in this study exhibited a range of physicochemical properties and variations in sulfur content in the Fujairah Port area. The study has provided increased knowledge and documentation of the differences in the physical and chemical properties of the VLSFOs from the Fujairah Port area.

The sample analysis standard set by the IMO for low-sulfur fuel is mainly designed for marine engines. The VLSFO oil used in marine transportation is highly variable; hence, the likely behavior of the VLSFOs if spilled at sea in the Arabian Gulf Region will also vary.

Further research will be needed to accurately define the typical ranges of density, viscosity, and other properties in VLSFOs. Additionally, more investigations are required to narrow these ranges based on the desulfurization methods used to produce the VLSFOs. According to verbal communication with shipowners in this region, they have encountered challenges since the introduction of the new VLSFO. The ship engines are not equipped to handle the different types of VLSFOs available in the region.

One recommendation for future research is to confirm the validity of the laboratories' sampling data by sending the same VLSFOs to multiple labs and then comparing their results with the two standards. If the data from multiple labs is statistically comparable, this will increase the ability to trust the laboratory sampling we receive and, therefore, strengthen our confidence in future analysis of the lab data.

The second recommendation is that the VLSFO samples should be tested at relevant sea temperatures. The temperature plays a vital role in the response to oil spills. The IMO has specified certain temperatures as standards for VLSFO, but these temperature ranges may differ for the heavier VLSFO. Understanding VLSFO behaviour when spilled at different sea temperatures is crucial.

Another recommendation is a joint study with refineries and shipowners in the Arabian Gulf area. The low sulfur content in oil is achieved in various ways in refineries, which can lead to significantly different oil properties among the VLSFOs. The study will help us further understand the fuel characteristics and the current challenges faced by VLSFO end-users, such as shipowners, port authorities, and oil spill responders.